Math for CS & AI: Homework 6

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Problem 3.12

Let $p(x)$ be a probability mass function supported on a finite set $\mathcal{X}$. That is, $p(x) > 0$ for all $x \in \mathcal{X}$. In information theory, the entropy of p is defined as

$$H(p) := \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{1}{p(x)}.$$

(a) Prove the following inequality:

$$0 \leq H(p) \leq \log_2 |\mathcal{X}|.$$

(b) Let $X_1, \ldots, X_n$ be n i.i.d. random variables drawn from p. For any $(x_1, \ldots, x_n) \in \mathcal{X}^n$, let $P(x_1, \ldots, x_n)$ be the probability of the event $\{X_1 = x_1, \ldots, X_n = x_n\}$. Prove that there exists $C > 0$ such that for all $\epsilon > 0$ and $n \geq 1$,

$$\Pr\left[2^{-n(H(p)+\epsilon)} \leq P(x_1, \ldots, x_n) \leq 2^{-n(H(p)-\epsilon)}\right] \geq 1 - 2e^{-C\epsilon^2 n}.$$

That is, when n is large, typically $P(X_1, \ldots, X_n) \approx 2^{-nH(p)}$. This is called the *asymptotic equipartition property* (AEP).

Answer:

For part (a), the function $f(x) = -x \log_2 x$, defined for $0 < x \leq 1$, is clearly non-negative, which implies $H(p) \geq 0$. Since the derivative of $f(x)$ is $f'(x) = -\log_2 x - \frac{1}{\ln 2}$, which is monotonically decreasing, we have that $f(x)$ is a concave function. Therefore, the maximum value of $H(p)$ is attained when $p(x) = \frac{1}{|\mathcal{X}|}$ for all $x \in \mathcal{X}$. Thus we can compute the maximum value of $H(p)$ as

$$\max H(p) = |\mathcal{X}| \cdot \frac{\log_2 |\mathcal{X}|}{|\mathcal{X}|} = \log_2 |\mathcal{X}|.$$

For part (b), let $f(x_1, \ldots, x_n) := \log_2 P(x_1, \ldots, x_n)$. Since $X_1, \ldots, X_n$ are i.i.d. random variables, we have:

$$\begin{aligned} f(x_1, \ldots, x_n) &= \log_2 P(x_1, \ldots, x_n) \\ &= \log_2 \left(\prod_{i=1}^{n} \Pr[X_i = x_i] \right) \\ &= \sum_{i=1}^{n} \log_2 p(x_i). \end{aligned}$$

Therefore, by the linearity of expectation, we have:

$$\begin{aligned}\mathbb{E}[f(x_1,\ldots,x_n)] &= \mathbb{E}\left[\sum_{i=1}^{n}\log_2 p(x_i)\right] \\ &= \mathbb{E}\left[-\sum_{i=1}^{n}\log_2\frac{1}{p(x_i)}\right] \\ &= -\sum_{i=1}^{n}\mathbb{E}\left[\log_2\frac{1}{p(x_i)}\right] \\ &= -nH(p).\end{aligned}$$

Let $\Delta = \max_{x\in\mathcal{X}}\log_2 p(x) - \min_{x\in\mathcal{X}}\log_2 p(x)$. Then for any fixed values of $x_1,\ldots,x_{i-1},x_{i+1},\ldots,x_n$ and any possible values $x_i, x_i' \in \mathcal{X}$, we have

$$|f(x_1,\ldots,x_i,\ldots,x_n) - f(x_1,\ldots,x_i',\ldots,x_n)| \le \Delta.$$

Applying McDiarmid's inequality, we have

$$\Pr\left[|f(x_1,\ldots,x_n) - \mathbb{E}[f(x_1,\ldots,x_n)]| \ge n\epsilon\right] \le 2e^{-2(n\epsilon)^2/L^2},$$

where $L^2 = n\cdot\Delta^2$. Rearranging the inequality, we obtain

$$\Pr\left[2^{-n(H(p)+\epsilon)} \le P\left(x_1,\ldots,x_n\right) \le 2^{-n(H(p)-\epsilon)}\right] \ge 1 - 2e^{-2\epsilon^2 n/\Delta^2} = 1 - 2e^{-C\epsilon^2 n},$$

where $C = \frac{2}{\Delta^2}$.

Problem 3.13

Let $p(x)$ be a probability mass function supported on a finite set $\mathcal{X}$. Alice randomly draws n i.i.d. random variables $X_1, \ldots, X_n$ from p and wants to send the values of these random variables to Bob. However, Alice's network is very slow, so she wants to minimize the number of bits she needs to send.

A *coding scheme* is a pair of encoding and decoding functions (f, g), where $f : \mathcal{X}^n \to \{0,1\}^*$ encodes the random variables $X_1, \ldots, X_n$ into a binary string $s \in \{0,1\}^*$ of arbitrary length, and $g : \{0,1\}^* \to \mathcal{X}^n$ decodes the binary string s back into the random variables $X_1, \ldots, X_n$.

A coding scheme is lossless if for all $(x_1, \ldots, x_n) \in \mathcal{X}^n$,

$$g\left(f\left(x_1, \ldots, x_n\right)\right) = \left(x_1, \ldots, x_n\right).$$

Design a lossless coding scheme such that

$$\mathbb{E}\left[\left|f\left(X_1, \ldots, X_n\right)\right|\right] \leq nH(p) + \mathcal{O}\left(\sqrt{n \log n}\right), \quad \text{as } n \to \infty,$$

where $|f(X_1, \ldots, X_n)|$ is the length of the binary string $f(X_1, \ldots, X_n)$.

Answer:

For any sequence $(x_1, \ldots, x_n)$ in the ϵ-typical set $A_\epsilon^{(n)}$, we have:

$$P(x_1, \ldots, x_n) \geq 2^{-n(H(p)+\epsilon)}.$$

Since the sum of the probabilities of all elements in the ϵ-typical set is at most 1, we find that the size of the ϵ-typical set is at most $2^{n(H(p)+\epsilon)}$. Therefore, by enumerating all elements in the ϵ-typical set in a fixed order, we can encode each sequence by its index in this enumeration, which requires $\lceil n(H(p)+\epsilon) \rceil = n(H(p)+\epsilon)+\mathcal{O}(1)$ bits.

Let $\epsilon = \sqrt{\frac{\ln n}{Cn}}$, where $C > 0$ is the constant derived in Problem 3.12. With this choice, we can encode sequences in the typical set using $nH(p) + \mathcal{O}\left(\sqrt{n \log n}\right)$ bits.

For sequences that are not in the typical set, we use a brute-force encoding: assign a unique $\lceil \log_2 |\mathcal{X}| \rceil$-bit binary code to each element of $\mathcal{X}$, then concatenate the codes for each element in the sequence, which uses $n\lceil \log_2 |\mathcal{X}| \rceil$ bits in total.

We prepend a 1-bit prefix to all encoded sequences to distinguish the two encoding types: a leading 0 indicates a typical-set index code, and a leading 1 indicates the brute-force symbol-by-symbol code.

For non-typical sequences, which occur with total probability at most $2e^{-C\epsilon^2 n} = \frac{2}{n}$, their contribution to the expected code length is at most $\frac{2}{n} \cdot n\lceil \log_2 |\mathcal{X}| \rceil = \mathcal{O}(1)$, which is negligible in the asymptotic limit. The 1-bit prefix adds a fixed 1 bit to all code lengths, which is also $\mathcal{O}(1)$. Therefore, the coding scheme satisfies

$$\mathbb{E}\left[\left|f\left(X_1, \ldots, X_n\right)\right|\right] \leq nH(p) + \mathcal{O}\left(\sqrt{n \log n}\right),$$

as $n \to \infty$.

Problem 3.14

Prove the *data processing inequality* for entropy: if X is a random variable with probability mass function p_X, and $Y = f(X)$ is a function of X, then the probability mass function of Y, denoted as p_Y, satisfies

$$H(p_Y) \leq H(p_X).$$

Then, use the data processing inequality to prove that in the setting of the previous problem, it is impossible to design a lossless coding scheme (f, g) such that

$$\mathbb{E}\left[|f\left(X_1, \ldots, X_n\right)|\right] < nH(p) - \log_2\left(2 + n\log_2|\mathcal{X}|\right), \quad \text{as } n \to \infty.$$

Answer:

For the first part, given $Y = f(X)$, for each possible value y of Y, let $S_y = \{x : f(x) = y\}$ be the set of all values of X that map to y. We compare the contributions to entropy from both sides. As shown in Problem 3.12, the function $g(x) = -x\log_2 x$ is concave, which by Jensen's inequality implies

$$-\left(\sum_{x\in S_y} p_X(x)\right)\log_2\left(\sum_{x\in S_y} p_X(x)\right) \leq -\sum_{x\in S_y} p_X(x)\log_2 p_X(x).$$

Summing over all possible values y of Y, we obtain

$$H(p_Y) \leq H(p_X).$$

We now consider the second part. Let $\boldsymbol{X} = (X_1, \ldots, X_n)$ denote the source sequence, $C = f(X_1, \ldots, X_n)$ denote the encoded binary string, and $\ell = |C|$ denote the length of the codeword.

First, if there exists any sequence $\boldsymbol{x} \in \mathcal{X}^n$ such that $|f(\boldsymbol{x})| > \lceil n\log_2|\mathcal{X}|\rceil$, then the coding scheme (f, g) cannot be optimal. This is derived by the fact that there are at most $|\mathcal{X}|^n = 2^{n\log_2|\mathcal{X}|}$ distinct sequences, so there must exist unused binary codes of length at most $\lceil n\log_2|\mathcal{X}|\rceil$ that can replace the longer codewords to reduce the expected length. Thus we may assume without loss of generality that all code lengths satisfy $\ell \leq \lceil n\log_2|\mathcal{X}|\rceil \leq n\log_2|\mathcal{X}| + 1$.

Second, we bound the entropy of the code length ℓ. Since ℓ takes at most $\lceil n\log_2|\mathcal{X}|\rceil + 1$ distinct values, by the upper bound on entropy proved in Problem 3.12, we obtain:

$$H(p_\ell) \leq \log_2\left(1 + \lceil n\log_2|\mathcal{X}|\rceil\right) \leq \log_2\left(2 + n\log_2|\mathcal{X}|\right).$$

Third, we compute the entropy of the source sequence $\boldsymbol{X} = (X_1, \ldots, X_n)$:

$$\begin{aligned}
H(p_{\boldsymbol{X}}) &= -\sum_{\boldsymbol{x}\in\mathcal{X}^n} \Pr[\boldsymbol{X} = \boldsymbol{x}]\log_2\Pr[\boldsymbol{X} = \boldsymbol{x}] \\
&= -\sum_{(x_i)\in\mathcal{X}^n}\left(\prod_{i=1}^n p(x_i)\right)\cdot\left(\sum_{i=1}^n \log_2 p(x_i)\right) \\
&= -\sum_{i=1}^n \sum_{(x_i)\in\mathcal{X}^n} p(x_i)\log_2 p(x_i)\prod_{\substack{1\leq j\leq n\\ j\neq i}} p(x_j) \\
&= nH(p).
\end{aligned}$$

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Lastly, let $L = \mathbb{E}[\ell]$ denote the expected code length. We expand the entropy of the encoded string C by grouping codewords by their length:

$$\begin{aligned}
H(p_C) &= -\sum_{c} \Pr[C=c] \log_2 \Pr[C=c] \\
&= -\sum_{i} \sum_{c:|c|=i} \Pr[C=c \mid \ell=i] \Pr[\ell=i] \left(\log_2 \Pr[C=c \mid \ell=i] + \log_2 \Pr[\ell=i]\right) \\
&= -\sum_{i} \Pr[\ell=i] \log_2 \Pr[\ell=i] \sum_{c:|c|=i} \Pr[C=c \mid \ell=i] \\
&\quad -\sum_{i} \Pr[\ell=i] \sum_{c:|c|=i} \Pr[C=c \mid \ell=i] \log_2 \Pr[C=c \mid \ell=i].
\end{aligned}$$

The first term equals $H(p_\ell)$, since $\sum_{c:|c|=i} \Pr[C=c \mid \ell=i] = 1$. The second term is the conditional entropy of C given $\ell = i$; since there are at most 2^i distinct binary strings of length i, this conditional entropy is at most $\log_2 2^i = i$. Therefore,

$$H(p_C) \leq H(p_\ell) + \sum_{i} \Pr[\ell=i] \cdot i = H(p_\ell) + L \leq \log_2 \left(2 + n \log_2 |\mathcal{X}|\right) + L.$$

Since the coding scheme is lossless, $\boldsymbol{X} = g(C)$, meaning the source sequence is a deterministic function of the encoded string. By the data processing inequality proved in the first part,

$$H(p_{\boldsymbol{X}}) \leq H(p_C).$$

Combining the inequalities above, we have

$$nH(p) \leq L + \log_2 \left(2 + n \log_2 |\mathcal{X}|\right),$$

which rearranges to

$$\mathbb{E}\left[|f\left(X_1, \ldots, X_n\right)|\right] \geq nH(p) - \log_2 \left(2 + n \log_2 |\mathcal{X}|\right).$$

This proves that the given inequality cannot be satisfied by any lossless coding scheme.

Problem 4.1

Consider the tiling of a $3 \times n$ rectangle using 1×2 dominoes, where n is an even number. Let a_n be the number of ways to tile the rectangle. Note that $a_0 = 1$ and $a_2 = 3$.

(a) Determine a closed-form formula for the generating function of a_n.

$$A(x) = \sum_{n \text{ even}} a_n x^n$$

(b) Determine a closed-form formula for a_n.

Answer:

For part (a), we inspect the right end of the rectangle. If there is no vertical domino in the last column, then there must be three horizontal dominoes covering the last two columns, and the number of tilings is a_{n-2}. Otherwise, there must be one vertical domino paired with horizontal dominoes; the vertical domino must align with another vertical domino an even number of columns away, with the space between filled by horizontal dominoes, giving 2 additional configurations for each smaller even-length tiling. Therefore, for any even $n \geq 2$:

$$a_n = a_{n-2} + 2 \sum_{0 \leq 2i < n} a_{2i}.$$

Substituting into the generating function, we obtain:

$$\begin{aligned} A(x) &= \sum_{\substack{n \geq 0 \\ n \text{ even}}} a_n x^n \\ &= 1 + \sum_{\substack{n \geq 2 \\ n \text{ even}}} x^n \left(a_{n-2} + 2 \sum_{0 \leq 2i < n} a_{2i} \right) \\ &= 1 + x^2 A(x) + \frac{2x^2}{1 - x^2} A(x). \end{aligned}$$

Solving the equation, we obtain

$$A(x) = \frac{1 - x^2}{1 - 4x^2 + x^4}.$$

For part (b), let φ and ψ be the two roots of the quadratic equation $y^2 - 4y + 1 = 0$, obtained by substituting $y = x^2$ into the denominator of $A(x)$, where

$$\varphi = 2 + \sqrt{3}, \quad \psi = 2 - \sqrt{3}.$$

Then we can decompose $A(x)$ into partial fractions:

$$A(x) = (1 - x^2) \left(\frac{\alpha}{1 - \varphi x^2} + \frac{\beta}{1 - \psi x^2} \right),$$

where α and β are constants determined by equating coefficients in $\alpha(1 - \psi x^2) + \beta(1 - \varphi x^2) = 1$:

$$\alpha = \frac{\sqrt{3} + 2}{2\sqrt{3}}, \quad \beta = \frac{\sqrt{3} - 2}{2\sqrt{3}}.$$

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Therefore, letting $n = 2t$ for $t \geq 0$, we can express the coefficient $a_n = a_{2t}$ as:

$$\begin{aligned} a_n &= \alpha\varphi^t + \beta\psi^t - \alpha\varphi^{t-1} - \beta\psi^{t-1} \\ &= \frac{1}{2\sqrt{3}}\left[\left(2+\sqrt{3}\right)^t \cdot \left(1+\sqrt{3}\right) - \left(2-\sqrt{3}\right)^t \cdot \left(1-\sqrt{3}\right)\right]. \end{aligned}$$